

It is hard at this date to write anything new on the subject of elementary geometry, and for the class addressed by the author it is not desirable, but the well-known facts may be treated in very diverse ways: in this case there is a novelty and freshness which must commend the treatment of them to all intelligent students. Take this "precise definition" of a plane: Take two points A and B and suppose two equal spherical bubbles formed about A and B as centres. Let them expand, always equal to each other, until they meet, and still keep on expanding. The line where the equal spherical bubbles, regarded as surfaces, meet, has all its points just as far from A as from B. As the bubbles still expand, this line, with all its points equidistant from A and B, itself expands and traces out a *plane* as its path through space. Hence we may define the *plane* as the region (or surface) where a point may be, that is, equidistant from two fixed points. . . It is evident that the plane, as thus defined, is reversible. . . The superiority of this definition consists in its not only telling what surface the plane is, but also making clear that there actually *is* such a surface. Thence our author readily derives the notion of the *ray* (anglicè, straight line: a *tract* being a part bounded by end-points).

This manner of illustration occurs repeatedly, and adds, we think, much to the interest of the book.

As a specimen of the mode of proof employed we take what is equivalent to part of Euc. i. 5. *Data*. ABC, an isosceles triangle, AB its base, AC and BC its equal sides (here we may remark the figure is badly drawn: a similar remark applies to figures on pp. 60 and 91). *Proof*. Take up the triangle ABC, turn it over, and replace it in the position BCA. Then the two triangles ACB and BCA have the equal vertical angles, C and C, also the side AC = BC (why?) and BC = AC (why?); hence they are congruent (why?), and the $\angle A = \angle B$.

In the more elaborate proofs there is a larger crop of "whys," and in some cases the interrogation is answered by the author.

The amount of ground covered is considerable, and yet, as we have gone through the whole of the text, it is so clearly opened up that the intelligent student, to whom we have previously referred, should be able to master it all, and successfully grapple with the well-chosen exercises which are arranged in fitting places throughout the book. "These exercises have been chosen with especial reference not so much to their merely disciplinary as to their didactic value, the author being persuaded that quite as good exercise may be found in going somewhat in walking round the square."

We have no hesitation in heartily commending Prof. Smith's introduction to teachers and pupils as an excellent one, and this we vouch, adapting the language of the learned counsel cited by Bailie Littlejohn, *nostro periculo*.

Primer of Horticulture. By J. Wright. (London: Macmillan and Co., 1893.)

THIS primer contains the substance of ten lectures delivered by Mr. Wright for the Surrey County Council. Besides the lectures, some sets of questions, asked after the lectures, are given together with the answers to these questions.

The primer is eminently practical, and is sure to prove very useful both to gardeners and to students. It cannot, however, be considered quite free from errors, and a careful revision would increase its value.

Sometimes the text is rather loose.

On p. 54 the word *pistil* is used indefinitely, sometimes meaning the style and at others the whole gynoecium.

Speaking of phosphatic manures on p. 64 the author says:—

"Mineral superphosphate is ground coprolite treated with sulphuric acid.

"Coprolite is antediluvian petrified manure, of which

there are large beds in the Eastern Counties. It is fairly active, yet sustaining.

"Thomas's phosphate powder, or basic slag— . . . is composed of 15 to 25 per cent. of phosphoric acid and about 45 per cent. of lime. It is not very quick in action, but lasting in effect."

From this description one cannot get much idea of the relative values of these three phosphatic manures, and basic slag suffers by comparison with ground coprolite. Practical experience shows that basic slag has a much higher value than ground coprolite as a manure, and has, moreover, an additional value as a check upon wireworm.

Again, on p. 77, the description of the fungus causing potato disease (*phytophthora infestans*) is scarcely accurate. In describing the aerial hyphae which spring from the mycelium in the leaves and push their way through the stomata, the author says:—

"These stem-growths of the fungus produce 'fruit'-spores (DD) in cells (Oogonia), that divide (F) and liberate the active agents in reproduction, tailed zoospores (G) which float in the air, and swim in the moisture, dew, or rain, on potato leaves." The letters in parentheses refer to fig. 18, p. 79. Neither text nor description below fig. 18 is correct. What Mr. Wright calls oogonia are really conidia, and what he calls conidia (F in fig. 18) represent the formation of an oospore from oogonium and antheridium. We must also dissent from the author's views on zoospores floating in the air.

Apart from these defects the primer is well worthy of perusal, and will no doubt meet with success. The practical part is very well done, and this is, of course, the most essential part of the book. WALTER THORP.

Ornithology in relation to Agriculture and Horticulture. Edited by John Watson. 220 pp. (London: W. H. Allen, 1893.)

THIS book contains a series of papers by well-known writers. The chief interest will gather around chapters iii. to vii. inclusive, which treat of the common sparrow. The trial of the sparrow is opened very ably by Mr. Chas. Whitehead (for the prosecution). He is well supported in the next chapter by Miss Eleanor A. Ormerod. These two writers for the prosecution will have the support of the vast majority of agriculturists in England, and their arguments contrast favourably with the less practical defence put forward in the two succeeding chapters by the Rev. F. C. Morris and the Rev. Theodore Wood.

Chapter VII. is written by J. H. Gurney, Jun., and from the result of 755 dissections he draws up a table showing that "in England about 75 per cent. of an adult sparrow's food is corn, chiefly barley and wheat, with a fair quantity of oats." Nobody with experience of grain-growing in England will deny that the sparrow is a terrible pest, and it is time that a movement be made not towards exterminating the troublesome bird, but towards reducing its numbers to normal limits.

Chapter IX. is an interesting defence of the rook, much of which defence this bird merits. It is written by O. V. Aplin, who also contributes a very useful chapter on miscellaneous small birds. WALTER THORP.

LETTERS TO THE EDITOR.

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Vectors versus Quaternions.

HAVING a vivid recollection of the pleasure I derived from Prof. Gibbs's attacks upon the quaternionic system in the rather one-sided discussion that took place about two years ago in this

journal, I have delayed replying to the letters of Profs. MacAulay and Tait, from an expectation that Prof. Gibbs would have something to say. In this I have not been mistaken; and, as there is a general agreement between us on the whole, I have merely to add some supplementary remarks. Prof. MacAulay refers to me as having raised the question again. I can assure him it has never been dropped. Apart from the one-sided discussion, it has been a live question with Prof. Gibbs and myself since about 1882, and is now more alive than ever. I cannot help thinking that Prof. MacAulay's letter was overhastily written, and feel sure that if he knew as much about the views and methods of those to whom he appeals as he does about Quaternions, he would have written it somewhat differently, or perhaps not have written it at all, from a conviction of the uselessness of his appeal. There is no question of suicide with us; on the contrary, quite the reverse. I am asked whether the "spoonfeeding," as he terms it, of Maxwell, Fitzgerald, &c., is not good enough for me. Why, of course not. It is quaternionic, and that is the real point concerned. Again, he thinks nothing of the inscrutable negativity of the square of a vector in Quaternions; here, again, is the root of the evil. As regards a uniformity of notation amongst antiquaternionists, I dare say that will come in time, but the proposal is premature. We have first to get people to study the matter and think about it. I have developed my system, such as it is, quite independently of Prof. Gibbs. Nevertheless, I would willingly adopt his notation (as I have adopted his dyadical notion of the linear operator) if I found it better. But I do not. I have been particularly careful in my notation to harmonise as closely as possible with ordinary mathematical ideas, processes, and notation; I do not think Gibbs has succeeded so well. But that matters little now; the really important thing is to depose the quaternion from the masterful position it has so long usurped, whereby the diffusion of vector analysis has been so lamentably impeded. I have been, until lately, very tender and merciful towards quaternionic fads, thinking it possible that Prof. Tait might modify his obstructive attitude. But there is seemingly no chance of that. Whether this be so or not, I think it is practically certain that there is no chance whatever for Quaternions as a practical system of mathematics for the use of physicists. How is it possible, when it is so utterly discordant with physical notions, besides being at variance with common mathematics? A vector is not a quaternion; it never was, and never will be, and its square is not negative; the supposed proofs are perfectly rotten at the core. Vector-analysis should have a purely vectorial basis, and the quaternion will then, if wanted at all, merely come in as an occasional auxiliary, as a special kind of operator. It is to Prof. Tait's devotion to his master that we should look for the reason of the little progress made in the last 20 years in spreading vector-analysis.

Now I have, in my turn, an appeal to make to Prof. MacAulay. I have been much interested in his recent R. S. paper. As the heart knoweth its own wickedness, he will not be surprised when I say that I seem to see in his mathematical powers the "promise and potency" of much future valuable work of a hard-headed kind. This being so, I think it a great pity that he should waste his talents on such an anomaly as the quaternionic system of vector analysis. I have examined a good deal of his paper, and can find nothing quaternionic about it except the language concerned in his symbols. On conversion to purely vectorial form, I find that it is greatly improved. I would suggest that he give up the quaternion. If he does not like my notation, or Prof. Gibbs's, or Prof. Macfarlane's, and will invent one for himself, it will receive proper consideration. He will greatly extend the sphere of his usefulness by the conversion. A difficulty in the way is that he has got used to quaternions. I know what it is, as I was in the quaternionic slough myself once. But I made an effort, and recovered myself, and have little doubt that Prof. MacAulay can do the same.

Passing to Prof. Tait's letter, it seems to be very significant. The quaternionic calm and peace have been disturbed. There is confusion in the quaternionic citadel; alarms and excursions, and hurling of stones and pouring of boiling water upon the invading host. What else is the meaning of his letter, and more especially of the concluding paragraph? But the worm may turn; and turn the tables.

It would appear that Prof. Tait, being unable to bring his massive intellect to understand my vectors, or Gibbs's, or Macfarlane's, has delegated to Prof. Knott the task of examining them, apparently just upon the remote chance that there

might possibly be something in them that was not utterly despicable. Prof. Knott has examined them, and has made some remarkable discoveries. One of them is that those vector methods in which the quaternion is not the master lead to formulæ of the most prodigious and alarming complexity. He has counted up the number of symbols in certain equations. Admirable critic!

Now, since this discovery, and Prof. Tait's remarks, are calculated to discourage learners, I beg leave to say, distinctly and emphatically, that there is no foundation for the imputation. Prof. Knott seems to have found a mare's nest of the first magnitude; unless, indeed, he is a practical joker, and has been hoaxing his venerated friend. Speaking from a personal knowledge of the quaternionic formulæ of mathematical physics, and of the corresponding formulæ in my notation and in Prof. Gibbs's, I can say definitely that there is very little to choose between them, so far as mere length goes. Perhaps Prof. Knott has been counting the symbols in a Cartesian formula, or in a semi cartesian one, or some kind of expanded form. I do not write for experts who delight in the most condensed symbolism. I do not even claim to be an expert myself. I have to make my readers, and therefore frequently, of set purpose, give expanded forms rather than the most condensed.

But so far as regards the brief vector formulæ, I find that the advantage is actually in my favour. I attach no importance to this, but state it merely as a fact which upsets Profs. Knott and Tait's conclusions. It is desirable that I should point out the reason, otherwise the fact may not be believed. In common algebra there is but one kind of product of a pair of quantities, say F and v , which is denoted by Fv . In vector algebra there are two kinds of products. One of these closely resembles the usual product, whilst the other is widely different, being a vector itself. Accordingly, to harmonise with common algebra, I denote the scalar product by $\mathbf{Fv}$. It degenerates to Fv when the vectors have the same direction. Now, since the quaternionists denote this function by $-\mathbf{SFv}$, which is double as long, whilst $\pm \mathbf{Fv}$ becomes $\mp \mathbf{SFv}$, it is clear that there must be an appreciable saving of space from this cause alone, because the scalar product is usually the most frequently occurring function.

But there are other causes. The quaternionic ways of specialising formulæ are sometimes both hard to read and lengthy in execution. Look at $S. UaUpS. U\beta Up$, which I see in Tait's book. I denote this by $(\alpha_1\rho_1)(\beta_1\rho_1)$, or else by $\alpha_1\rho_1 \cdot \beta_1\rho_1$. Tait is twice as long. But the mere shortness is not important. It is distinctness that should be aimed at, and that is also secured by departing from quaternionic usage. Examples of shortening and clarifying by adopting my notation may be found on nearly every page of Tait's book.

Consider, for example, rotations. Quaternionists, I believe, rather pride themselves upon their power of representing a rotation by means of a quaternion. Thus, $\mathbf{b} = q\mathbf{a}q^{-1}$. The continued product of a quaternion q , a vector $\mathbf{a}$, and another quaternion q^{-1} , produces a vector $\mathbf{b}$, which is $\mathbf{a}$ turned round a certain axis through a certain angle. It is striking that it should turn out so; but is it not also a very clumsy way of representing a rotation, to have to use two quaternions, one to pull and the other to push, in order to turn round the vector lodged between them? Is it not plainer to say $\mathbf{b} = r\mathbf{a}$, where r is the rotator? Then we shall have $\mathbf{ac} = ar'r'\mathbf{c} = r'\mathbf{a}r'\mathbf{c}$ &c., if r' is the reciprocal of r . Then Prof. Tait's $Vq\mathbf{a}q^{-1}q\phi(q^{-1}\mathbf{b})q^{-1}$ is represented by $Vr\mathbf{a}r\phi r'\mathbf{b}$. See his treatise, p. 326, 3rd edition, and note how badly the $q(\)q^{-1}$ system works out there and in the neighbouring pages.

What, then, is this rotator? It is simply a linear operator, like ϕ . It is, however, of a special kind, since its conjugate and its reciprocal are one, thus $rr' = 1$, or $r' = r^{-1}$. Far be it from me to follow Prof. Tait's example (see his letter) and impute to him an "imperfect assimilation" of the linear and vector operator. What I should prefer to suggest is that his admiration for the quaternionic mantle is so extreme that he will wear it in preference to a better-fitting and neater garment. If we like we can express the rotator in terms of a quaternion, in another way than above, though involving direct operations only. But I am here merely illustrating the clumsiness of the quaternionic formulæ in physical investigations, and their unnaturalness, by way of emphasising my denial and disproof of the charge made by Prof. Tait against vectorial methods. The general anti-quaternionic question I have considered elsewhere.

Paignton, Devon, March 24.

OLIVER HEAVISIDE.